Data Appendix

Classifying Leukemic B-Lymphoblast Cells from Normal B-Lymphoid Precursors Using Deep Learning on Microscopic Blood Smear Images

Group: Model Citizens | Shaina Banduri, Neil Parikh, Nishana Dahal | DS 4002

1. Dataset Description

The C-NMC 2019 (Cancer vs. Normal Microscopic Cell) dataset was obtained from The Cancer Imaging Archive at the National Cancer Institute. It contains labeled microscopic blood smear cell images collected from 118 patients (69 cancer subjects, 49 normal subjects). Images were stained using the Jenner-Giemsa protocol and captured at 450×450 pixel resolution in BMP format. The dataset is pre-divided into three cross-validation folds for training and a separate held-out preliminary test set for final evaluation.

2. Unit of Observation

Each observation is a single microscopic blood smear cell image (one BMP file) associated with one patient. Each image carries a binary class label: 1 = leukemic B-lymphoblast (cancer / ALL) or 0 = normal B-lymphoid precursor (hem). The dataset contains no tabular covariates beyond the image file and its class label.

3. Summary Statistics

Split / Fold	Total Images	Cancer (all)	Normal (hem)
Training — all folds combined	10,661	7,272 (68.2%)	3,389 (31.8%)
fold_0	3,527	2,397 (67.9%)	1,130 (32.1%)
fold_1	3,581	2,418 (67.5%)	1,163 (32.5%)
fold_2 (validation split)	3,553	2,457 (69.2%)	1,096 (30.8%)
Test (held-out preliminary set)	1,867	1,219 (65.3%)	648 (34.7%)
Total	12,528	8,491 (67.8%)	4,037 (32.2%)

Note: A consistent ~2:1 class imbalance (cancer:normal) is present across all splits. This is addressed during training via inverse-frequency class weights in the cross-entropy loss.

4. Data Dictionary

The dataset is organized as a directory of BMP image files. Class membership and fold assignment are encoded in the folder structure. The preliminary test set additionally provides a CSV file (C-NMC_test_prelim_phase_data_labels.csv) mapping each image filename to its ground-truth label.

4a. Training Data — Folder-Structure Variables

Variable	Description	Type	Example Values
fold	Cross-validation fold.	Categorical	fold_0 / fold_1 / fold_2
class_folder	Subfolder encoding class label.	Categorical (binary)	all (cancer), hem (normal)
label	Numeric label derived from class_folder.	Binary int	1 = cancer, 0 = normal
filename	BMP image filename (encodes patient/slide	String	UID_57_29_1_all.bmp
image	Raw microscopic blood smear cell image.	BMP, 450×450 px	—

4b. Test Set — CSV Variables (C-NMC_test_prelim_phase_data_labels.csv)

Variable	Description	Type	Example Values
Patient_ID	Original patient-level filename (encodes cla	String	UID_57_29_1_all.bmp
new_names	Simplified numeric filename; used to locate	String	1.bmp, 4.bmp
labels	Ground-truth binary class label.	Binary int	1 = cancer, 0 = normal

4c. Dataset Metadata

Field	Description	Type	Value
Cancer type	Type of cancer represented in positive cl	Constant	Acute Lymphoblastic Leukemia (ALL
Cancer location	Anatomical site.	Constant	Blood & Bone Marrow
Staining protocol	Slide preparation procedure.	Constant	Jenner-Giemsa
Native resolution	Raw BMP image size.	Integer (px)	450 × 450
Model input size	Resolution after preprocessing resize.	Integer (px)	224 × 224
Normalization	Per-channel pixel normalization statistics	Float	ImageNet mean/std
Source	Dataset provider.	String	The Cancer Imaging Archive (TCIA),
Last updated	Most recent dataset release date.	Date	2019-05-28

5. Current Unknowns

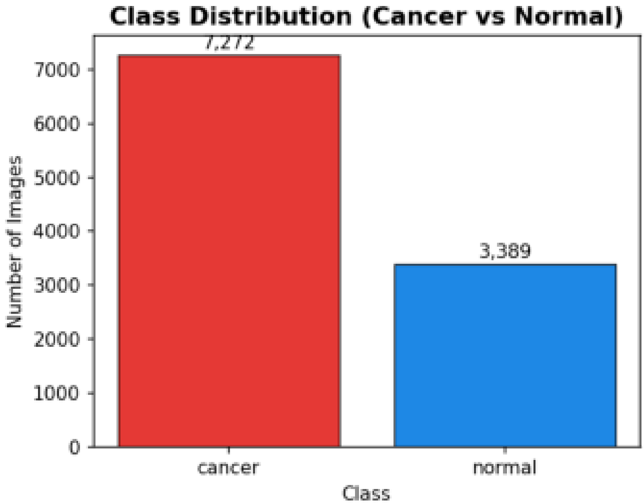
It is unknown whether image characteristics differ systematically across patients or slides. Individual patients may exhibit unique visual signatures that could lead the model to learn patient-specific patterns instead of disease-relevant features. The fold-based split partially mitigates this risk by grouping images from the same patient into the same fold, but patient-level leakage cannot be fully ruled out without explicit patient-stratified cross-validation.

6. Exploratory Data Analysis

Three questions were explored during EDA. Figures are reproduced directly from output/.

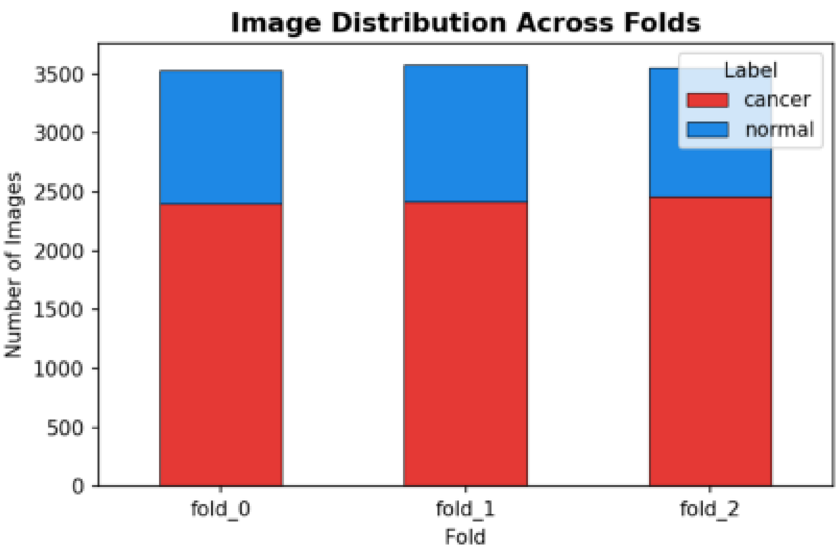
Q1: Is the training dataset class-balanced?

No. The training set contains 7,272 cancer images (68.2%) and 3,389 normal images (31.8%), yielding an approximate 2:1 cancer-to-normal ratio that is consistent across all three folds. Inverse-frequency class weights were applied during training to compensate.



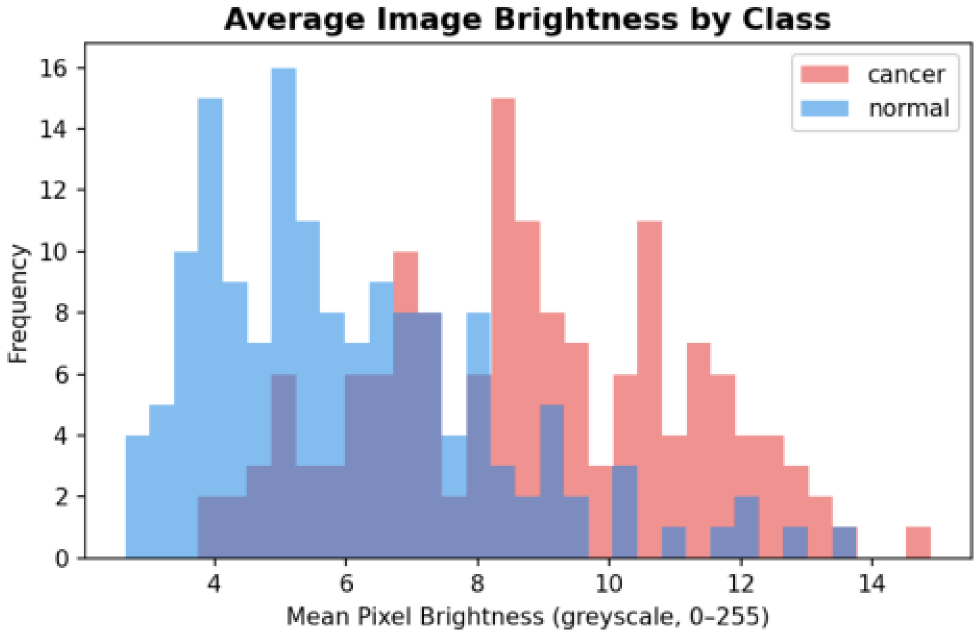
Q2: How are images distributed across folds?

Images are evenly distributed: fold_0 (n=3,527), fold_1 (n=3,581), fold_2 (n=3,553). The ~2:1 class imbalance is consistent within each fold, confirming a stratified split. fold_0 and fold_1 were used for training; fold_2 served as the validation set.



Q3: Do cancer and normal images differ in average brightness?

Yes. A sample of 150 images per class (300 total) was used to compare mean greyscale pixel brightness. Cancer images exhibit higher average brightness and a wider spread (roughly 7–15), while normal images are concentrated at lower brightness levels (roughly 3–8). This suggests greater stain uptake or morphological contrast in leukemic cells, consistent with known pathological characteristics of ALL.



7. References

[1] American Cancer Society, "Key statistics for childhood leukemia," 2023. [Online]. Available: https://www.cancer.org/cancer/types/leukemia-in-children/key-statistics.html [Accessed: Mar. 30, 2026].

[2] "C-NMC-2019," The Cancer Imaging Archive (TCIA). https://www.cancerimagingarchive.net/collection/c-nmc-2019/

[3] S. J. Pan and Q. Yang, "A survey on transfer learning," IEEE Trans. Knowl. Data Eng., vol. 22, no. 10, pp. 1345–1359, Oct. 2010.